

Simanta Lahkar

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EDUCATION

PhD	Shanghai Jiao Tong University, Materials Science and Engineering Dissertation: “The effects of structural anisotropy on mechanical properties of hexagonal boron nitride and uncertainty in characterization.” Advisor: Dr. Kolan Madhav Reddy (Associate Professor).	6/2022
MSc + BE	BITS Pilani, Physics (honors) + Mechanical Engineering (honors) CGPA: 8.21/10 (dual degree program). Special Project: “Atomistic modeling of mixed-mode fracture in amorphous silica.” Advisor: Dr. Gaurav Singh (Assistant Professor).	7/2018

RESEARCH EXPERIENCE

Indian Institute of Technology – Gandhingar, GJ, India 9/2022 – present
Early Career Fellow, Discipline of Materials Engineering.

Recruited as an independent researcher by the institute to work with a Faculty collaborator on a self-proposed project and other collaborations, as well as teach courses. Supervised undergraduate and graduate students’ research projects.

Key research contributions:

- Determining thermo-mechanical properties of defect-engineered materials using simulations and data-driven methods in order to obtain high performance designs.
- Creation of python scripts to make atomistic models of 2D materials with realistic grain boundary structures using CVT algorithm.
- Mapping of the phonon modes in high entropy alloys (HEA) and 2D materials by analyzing molecular dynamics (MD) simulations data.
- Investigating 2D materials by first-principles calculations for nitrogen reduction applications.

Shanghai Jiao Tong University, SH, China 4/2019 – 7/2022
Graduate Research Assistant, School of Materials Science and Engineering.

- Characterized structure and failure behavior of inorganic solids using electron microscopy and spectroscopy, Raman, XRD analysis, & computational modeling methods.
- Performed MD simulations of 2D materials and discovered an intrinsic distortion-mediated toughening and strengthening mechanism in hBN.
- Devised a strategy to optimize Tsai-Wu failure criterion for hBN in full 3D by performing strategic failure simulations, leading to over 99% accuracy.
- Formulated a framework for quantitatively analyzing the accuracy of any failure criterion in full 6D stress-space compared to test data.

- Developed a method to calculate core-loss EELS peak intensities, and verified the tilt-dependence of experimental spectra using theoretical relativistic scattering calculations.
- Verified ripple – formation in spark plasma sintered (SPS) hBN (+Fe) by TEM characterization.

Shanghai Jiao Tong University, SH, China

7/2017 - 3/2019

Research Assistant, Department of Engineering Mechanics, School of NAOCE.

- Studied electron irradiation of WSe₂ using MD simulations in order to reveal defect evolution.
- Investigated thermal transport in functionalized graphene using atomistic simulations in order to find the thermal-rectification mechanism.
- Determined fracture behavior of woven nano-fabrics using MD simulations.

AWARDS AND HONORS

- Best Oral Presentation Award – ICEFA 2022 by Elsevier (7/2022).
- Work featured on the front cover of *Mechanics of Materials* journal (7/2022).
- Outstanding Graduate Award – SJTU (12/2021).
- CSC Doctoral Scholarship (9/2018 to 6/2022).

TECHNICAL SKILLSET

Programming & packages: Python, LAMMPS, Ovito, CP2K, VESTA, Abaqus, Solidworks, Machine learning (tensorflow, scikit-learn, deepchem), Linux.

Experimental & analytical: SEM, TEM, EELS, EDS, XRD analysis, Raman Spectroscopy, XPS, UV-vis spectroscopy, CVD.

PROFESSIONAL AND TEACHING EXPERIENCE

Shanghai Jiao Tong University, SH, China

9/2020 - 1/2021

Teaching Assistant, *Structure of Materials -MT330*.

- Managing student homeworks and assignments for undergraduate course.

JV Institute, SH, China.

6/2020 - 11/2020

Project teacher (part-time)

- Taught offline interactive activity-focused physics classes to high school students.

Tezpur University, AS, India.

6/2017 - 7/2017

Summer Intern

- Used homology modeling to investigate unknown structure of several proteins.

Kudankulam Nuclear Power Plant, TN, India.

6/2015 - 8/2015

Summer intern

- Implemented a machine learning regression model from scratch to automate power estimation.

ADDITIONAL COURSE SPECIALIZATIONS

- Machine Learning in Molecular and Materials Sciences (UM – SJTU Joint Institute), Elastic-plastic Mechanics (SJTU), Fracture Mechanics (SJTU), Multiscale Materials Modeling and Simulation (SJTU), Machine Learning Crash Course (Google), Data Science with Python (Google).

OTHERS

- **Conference Speaker**, IXth International Conference on Engineering Failure and Analysis (ICEFA 2022), 7/2022. DOI: <http://dx.doi.org/10.13140/RG.2.2.22037.81120>
- **Conference Speaker**, International Conference on Frontier Materials (ICFM 2022), 3/2022. DOI: <https://dx.doi.org/10.12351/ks.2203.1044>
- Mentored undergraduate students for MD simulations - based summer project in BITS Pilani.
- Hobbies: Financial markets, drones, and photography.

PUBLICATIONS

Journal Papers

- Jeong B, [Lahkar S](#), An Q, Reddy KM (2022), "Mechanical properties and deformation behavior of superhard lightweight nanocrystalline ceramics". *Nanomaterials*, 12(18):3228
- Huo X, Jin M, Jin Q, [Lahkar S](#), Huang G, Lu W, Wang X (2022), "Mechanical response and actuation fatigue behavior of Nb-nanowire enhanced NiTiHf high temperature shape memory alloy wires". *Scripta Materialia*, 218: 114839.
- Karati A, [Lahkar S](#), Hu Y, Sahoo R, Adigilli HK, Hebalkar NY, Reddy KM, Joardar J (2022), "Cu-Mo-S Mixed Chalcogenide Nanostructures for Pseudocapacitive Applications". *ACS Applied Nano Materials*, 5(8):10203-12
- [Lahkar S](#) and Reddy KM (2022), "Atomistically calibrated Tsai-Wu criterion for orthotropic layered 2D materials and interpreting accuracy-A study on failure of hBN". *Mechanics of Materials*, 170: 104345.
-Featured on the Journal's Front Cover.
- [Lahkar S](#), Jeong B, Wang X, Hemker K, Reddy KM (2022), "Intrinsic strengthening and toughening in hexagonal boron nitride by ripples". *Acta Materialia*, 229: 117845.
- Karre R, Shen Y, Song S, Hu Y, [Lahkar S](#), Wang X, An Q, Reddy KM (2022), "Shear band formation during nanoindentation of EuB6 rare-earth hexaboride". *Communications Materials*, 3, 24.
- Huo X, Chen P, [Lahkar S](#), Jin M, Han X, Song Y, Wang X (2022), "Occurrence of the R-phase with increased stability induced by low temperature precipitate-free aging in a Ni50.9Ti49.1 alloy". *Acta Materialia*, 227: 117688.
- Zhang H, Örnek M, [Lahkar S](#), Song S, Wang X, Haber RA, Reddy KM (2022), "Enhanced densification and mechanical properties of β -boron by in-situ formed boron-rich oxide". *Journal of Materials Science & Technology*, 99: 148-60.
- [Lahkar S](#), Wang X, Hemker K, Reddy KM (2021), "Characterization and understanding of the tilt-dependence of core-loss spectra for hexagonal boron nitride". *Scripta Materialia*, 204: 2137-2144.
- Zhang H, Shen Y, [Lahkar S](#), Fujita T, Hemker K, An Q, Reddy KM (2021), "Distorted planar defects stabilize tetragonal boron". *Scripta Materialia*, 194: 113685.
- Wei A, [Lahkar S](#), Li X, Li S, Ye H (2019), "Multilayer graphene-based thermal rectifier with interlayer gradient functionalization". *ACS applied materials & interfaces*, 11(48): 45180-8.

- Reddy KM, Guo D, Lahkar S, Cheng C, Shinoda Y, An Q, Wang X (2019), “Graphite interface mediated grain-boundary sliding leads to enhanced mechanical properties of nanocrystalline silicon carbide”. *Materialia*, 7: 100394.
- Li Y, Lahkar S, Wei Q, Qiao P, Ye H (2019), “Strength nature of two-dimensional woven nanofabrics under biaxial tension”. *International Journal of Damage Mechanics*, 28 (3): 367-79.

Conference Paper

- Lahkar S, Ghosal S, Singh G, “Atomistic simulation of mixed mode fracture in vitreous silica”. *Proceedings of ASME 2017 International Mechanical Engineering Congress and Exposition*, Nov. 3-9, 2017, Tampa, Florida, USA.

Book Chapter

- Reddy KM, Lahkar S, Hirata A, Chen M (2020), “Atomic structure of Boron Carbide”. In Reddy KM, “Boron carbide: structure, processing, properties and applications”. *Nova Science Publishers*.

REFERENCES

(Reference #1: PhD Advisor at SJTU)

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(Reference #2: Chair of PhD Committee and close collaborator at SJTU)

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 Shanghai Jiao Tong University
 SMSE building A, Room 107, 800 Dongchuan Rd., Minhang, SH, China
 Phone: +86-137 0161 4094
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(Reference #3: Undergraduate project guide at BITS Pilani – Goa Campus)

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